# Recap: Lecture 23

In Lecture 23, we

- Applied SoV to a second problem involving the 1D heat equation and homogeneous BCs
- Solved a problem involving inhomogeneous BCs by writing

$$u(x,t) = u_S(x) + u_T(x,t)$$

where $u_S$ satisfies the inhomogeneous BCs and is time independent and $u_T$ satisfies homogeneous BCs

# Lecture 24

In Lecture 24, we will:

1. Consider 2D heat equation on circular panels
2. Apply SoV to an axisymmetric problem with homogeneous BCs
3. Find the steady state solution for a problem involving inhomogeneous BCs by applying SoV to Laplace's equation







https://upload.wikimedia.org/wikipedia/commons/5/59/SolarStirlingEngine.jpg





https://www.researchgate.net/figure/US-systems-22-kWel-SAIC-and-10-kWel-WGA_fig11_282859077

# Case study 8:
# Heat conduction in 2D

## Heat conduction in 2D: Problem 1

Find the evolution of the temperature of circular panel whose edge is held at zero temperature and subject to an axisymmetric initial condition.

- Laplace's equation in cylindrical polar coordinates for $u(\mathbf{x}, t) = u(r, t)$ is

$$\frac{\partial u}{\partial t} = \alpha \nabla^2 u \equiv \alpha \left( \frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} \right) \quad \text{for} \quad 0 < r < a, \;\; t > 0$$

- Boundary conditions:

$$u(r = a, t) = 0 \quad \text{for} \quad t > 0$$

$$\lim_{r \to 0} u(r, t) \text{ is finite (regularity)}$$

- Initial condition

$$u(r, t = 0) = f(r) \quad \text{for} \quad 0 < r < a$$

Seek solution using separation of variables

1. Separate variables to turn single PDE into multiple ODEs

$$u(r,t) = R(r)\,T(t)$$

$$\Rightarrow \quad R\,\dot{T} = \alpha(R'' + r^{-1}\,R')T \qquad \text{(PDE)}$$

$$\Rightarrow \quad \frac{\dot{T}}{\alpha\,T} = \frac{R'' + r^{-1}\,R'}{R} = \lambda$$

$$\Rightarrow \quad \dot{T} - \alpha\,\lambda\,T = 0 \quad \text{and} \quad R'' + r^{-1}\,R' - \lambda\,R = 0 \qquad \text{(two ODEs)}$$

We interpret the second equation as an eigenvalue equation for $\lambda$ and $R(r)$, with $\lambda$ constrained by the boundary conditions

2. Apply the boundary condition to find the eigenvalues and eigenfunctions

$$u(a,t) = 0 \quad \Rightarrow \quad R(a)\,T(t) = 0 \quad \Rightarrow \quad R(a) = 0$$

Let $\lambda = -k^2$, so that

$$R'' + r^{-1}\,R' + k^2\,R = 0$$

and

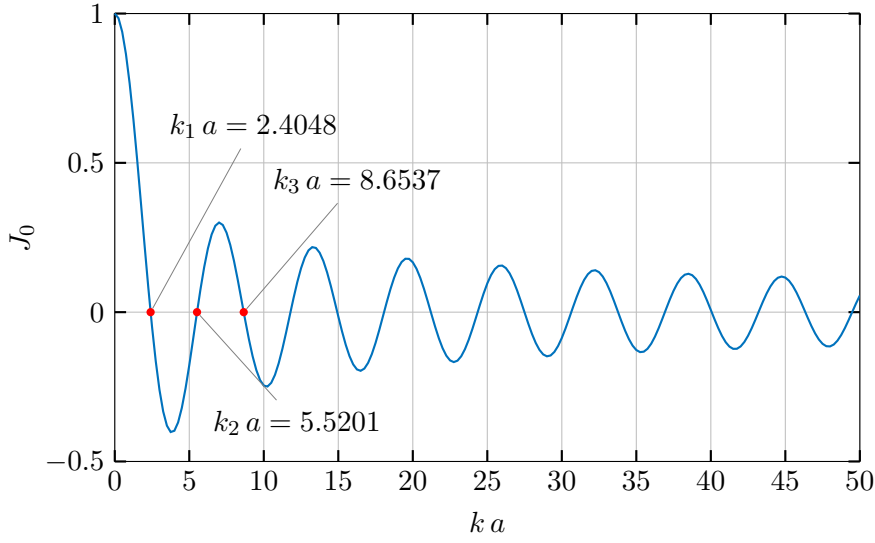
$$s = k\,r \quad \text{and} \quad S(s) = R(r) \quad \Rightarrow \quad R' = k\,S' \quad \Rightarrow \quad R'' = k^2\,S''$$

so that we have Bessel's equation of order zero (cf. Ex. 2.8)

$$S'' + \frac{1}{s}\,S' + S = 0 \quad \Rightarrow \quad S(s) = C_1\,J_0(s) + C_2\,y_2(s)$$

$$\Rightarrow \quad R(r) = A\,J_0(kr) + B\,y_2(kr)$$

We require a bounded solution as $r \to 0$, so $B = 0$. The BC gives the condition $J_0(ka) = 0$. There are infinitely many solutions $k_n$ as $J_0$ has infinitely many zeros in $\mathbb{R}_{>0}$ (see next page).


1
0.5
0
−0.5
$J_0$
$k_1\, a = 2.4048$
$k_3\, a = 8.6537$
$k_2\, a = 5.5201$
0
5
10
15
20
25
30
35
40
45
50
$k\, a$

3. Solve the remaining ODEs for each eigenvalue

$\lambda = -k_n^2$, so

$$\dot{T}_n + \alpha\, k_n^2\, T_n = 0$$

$$\Rightarrow \quad T_n(t) = \mathrm{e}^{-\alpha\, k_n^2\, t}$$

4. Apply superposition principle to get a series general solution

$$u(r,t) = \sum_{n=1}^{\infty} C_n \, \mathrm{e}^{-\alpha \, k_n^2 \, t} \, J_0(k_n r)$$

5. Apply the initial conditions to find the series coefficients

## Orthogonality condition for $J_0$

$$\int_0^a J_0(k_m r)\, J_0(k_n r)\; r\,\mathrm{d}r = \begin{cases} \dfrac{a^2}{2} J_1^2(k_n a) & (n = m) \\ 0 & (n \neq m) \end{cases}$$

The LHS may be interpreted as a scalar product in a radial coordinate. The RHS features another Bessel function $J_1$ which we do not discuss here.

More on all this in MAST30031 Methods of Mathematical Physics.

The IC is

$$u(r,0) = \sum_{n=1}^{\infty} C_n\, \mathrm{e}^{-\alpha\, k_n^2\, t}\, J_0(k_n r) = f(r)$$

Then

$$C_n = \frac{2}{a^2\, J_1^2(k_n a)} \int_0^a f(r)\, J_0(k_n r)\; r\,\mathrm{d}r \qquad (n = 1, 2, \ldots)$$

# Laplace's equation: Steady-state solutions of 2D heat equation

- The heat equation in 2D is
$$\frac{\partial u}{\partial t} = \alpha \nabla^2 u$$
- Any steady-state solution $u_S(\mathbf{x})$ is time-independent and hence needs to satisfy Laplace's equation,
$$\nabla^2 u_S = 0$$
- Decompose a general solution as
$$u(\mathbf{x}, t) = u_T(\mathbf{x}, t) + u_S(\mathbf{x}) \quad \text{such that} \quad u(\mathbf{x}, t) \to u_S(\mathbf{x}) \quad \text{as } t \to \infty$$
- We note that $u_S(\mathbf{x})$ is independent of the initial condition and only depends on the boundary conditions.
- We use the steady-state solution to satisfy inhomogeneous BCs

## Heat conduction in 2D: Problem 2

Find the steady state for a circular panel, subject to an inhomogeneous BC which is not necessarily axisymmetric.

- Laplace's equation in cylindrical polar coordinates for $u_S(\mathbf{x}) = u_S(r, \theta)$ is

$$\nabla^2 u_S \equiv \frac{\partial^2 u_S}{\partial r^2} + \frac{1}{r}\frac{\partial u_S}{\partial r} + \frac{1}{r^2}\frac{\partial^2 u_S}{\partial \theta^2} = 0 \quad \text{for} \quad 0 < r < a, \;\; 0 \leqslant \theta < 2\pi$$

- Inhomogeneous BC

$$u_S(r = a, \theta) = f(\theta) \quad \text{for} \quad 0 \leqslant \theta < 2\pi$$

Seek solution using an adapted version of separation of variables

1. Separate variables to turn single PDE into multiple ODEs

$$u_S(r,\theta) = R(r)\,\Theta(\theta)$$

$$\Rightarrow \quad R''\,\Theta + \frac{1}{r}R'\,\Theta + \frac{1}{r^2}\,R\,\Theta'' = 0 \quad \text{(PDE)}$$

$$\Rightarrow \quad \frac{R'' + r^{-1}\,R}{r^{-2}\,R} = -\frac{\Theta''}{\Theta} = \mu = \text{const}$$

$$\Rightarrow \quad \Theta'' + \mu\,\Theta = 0 \quad \text{and} \quad R'' + \frac{1}{r}\,R' - \frac{\mu}{r^2}\,R = 0 \qquad \text{(two ODEs)}$$

2. Find eigenvalues and eigenfunctions

Solve $\Theta'' + \mu\,\Theta = 0$, subject to periodic BCs $\Theta(2\pi) = \Theta(0)$ and $\Theta'(2\pi) = \Theta'(0)$.

Case $\mu = -k^2 < 0$

$$\Theta(\theta) = A\,\mathrm{e}^{k\theta} + B\,\mathrm{e}^{-k\theta}$$

which only satisfies the periodic BCs if $A = B = 0$, i.e., trivial solution.

Case $\mu = 0$

$$\Theta(\theta) = A\,\theta + B$$

for which the periodic BCs dictate that $A = 0$ and hence we find $\Theta(\theta) = B$.

Case $\mu = k^2 > 0$

$$\Theta(\theta) = A\,\cos(k\theta) + B\,\sin(k\theta)$$

and the periodic BCs require $k = n$, so that

$$\Theta(\theta) = A\,\cos(n\theta) + B\,\sin(n\theta)$$

3. Solve the remaining ODEs for each eigenvalue

For $\lambda = 0$

$$R'' + \frac{1}{r}R' = 0 \quad \Rightarrow \quad R = A + B\,\log(r)$$

and boundedness implies $B = 0$, so that $R = A$

For $\lambda = n^2$

$$R'' + \frac{1}{r}R' - \frac{n^2}{r^2}R = 0$$

We solved this ODE in Ex 1.8,

$$R(r) = A\,r^n + B\,r^{-n}$$

and boundedness implies $B = 0$, so that $R(r) = A\,r^n$

4. Apply Superposition Principle to get a series general solution

Combine every possible solution of the $\Theta$-ODE with every possible solution of the $R$-ODE (for the same eigenvalue $\mu$) and sum over all possible eigenvalues

$$u_S(r,\theta) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} r^n \left\{a_n \cos(n\theta) + b_n \sin(n\theta)\right\}$$

5. Apply the inhomogeneous boundary conditions to find the series coefficients

$$f(\theta) = u_S(a,\theta) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} a^n \left\{a_n \cos(n\theta) + b_n \sin(n\theta)\right\} \quad \text{for} \quad 0 \leqslant \theta < 2\pi$$

The RHS is the periodic extension of $f(\theta)$ so that

$$a^n a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(\theta)\cos(n\theta)\,\mathrm{d}\theta \quad \text{and} \quad a^n b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(\theta)\sin(n\theta)\,\mathrm{d}\theta$$